

Mantavya Mahajan

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EDUCATION

The Pennsylvania State University

B.S.: Computer Science, Mathematics, Business Minor

University Park, PA

Graduation: December 2025

Technical Skills

Programming languages: Python, Java, C, C++, TypeScript, JavaScript, SQL, Bash

Technologies: Node.js, Django, Express.js, Flask, GraphQL, PostgreSQL, MongoDB, MySQL, Snowflake

Software Development: Data Structures & Algorithms, OOP, System Design, Parallel Computing, ETL Pipelines

DevOps & Cloud: Docker, Kubernetes, AWS (Lambda, EC2, S3), CI/CD, Jenkins, Terraform, Azure, Hugging Face

Tools: Git/GitHub, Linux, JIRA, Redis, Celery, WebSocket's, Dask, Selenium

Scale AI

Remote, US

Technical Advisor Intern

May 2025 – Aug 2025

- Evaluated and optimized **large language model** outputs for code generation across multiple programming languages including **Python**, **C++** and **Typescript**, improving accuracy by 15%
- Developed evaluation frameworks and **benchmarks** using **Python** to assess **AI model performance** and training data quality across 500+ test cases
- Built **RAG system** at system level for log dump analysis, increasing task completion speeds by 30%

M8

Remote, US

Lead Software Developer

Aug 2024 – May 2025

- Developed efficient and scalable **MERN stack** APIs for **high concurrent user traffic**, optimizing **request routing and load balancing**, improving throughput by 20%
- Implemented **RBAC and OAuth authentication**, enhancing data security and user access control across services **deployed on AWS**
- Optimized **React/Redux** performance and created responsive UI components with **Material-UI**, reducing load times by 40%

Alchem International

Delhi, IN

Software Engineering Intern

Jan 2024 – Dec 2024

- Refactored legacy monolithic architecture** into scalable microservices using **Python** and **Django**, reducing system downtime by 25%
- Refactored PostgreSQL queries** and optimized indexing while improving **CI/CD pipelines** with **Python** scripts, boosting performance by 20%
- Optimized API codebase** and **React** frontend through multithreading, caching, and **code refactoring**, cutting response time by 30%

Projects

FinGuard (React Native, Node.js, MongoDB, Google Maps API, Gemini API)

Mar 2025

- Built **React Native mobile app** using **GPS location services** to identify nearby stores and provide financial insights
- Developed **location-based recommendation** with **MongoDB** and **Google Maps API** for optimal credit card suggestions
- Integrated **real-time merchant analysis** and scam detection alerts based on user location and store data

EPU Index Replication & NLP Pipeline (Python, Django, ProQuest API, NLP, Tableau)

Ongoing

- Developed **NLP pipeline** using **spaCy**, **NLTK**, **TF-IDF** to extract policy uncertainty metrics from articles
- Automated **ProQuest API ingestion** with **PostgreSQL** and **Django ORM** for data storage
- Designed **Tableau dashboards** with time-series forecasting for economic trend analysis

Meal Plan Optimizer (React, Firebase, TypeScript, Puppeteer, Express, PostgreSQL, Django)

Dec 2024

- Built **Django REST API** to analyze dining transactions and provide personalized budget insights
- Automated **web scraping** with **Puppeteer** and MFA authentication for transaction retrieval
- Integrated **Celery** and **Redis** for real-time meal spending tracking

Research Experience

Product Recommendation System

Jan 2024 – May 2024

- Implemented **BERT4Rec transformer model** for next-basket prediction on Instacart dataset, achieving **64% Hit@10** accuracy
- Built **preprocessing pipeline** handling **130K+ customer sequences** with masked language modeling
- Developed **clustering analysis** using **t-SNE** and **adaptive HDBSCAN** to validate model embeddings